

1 · THE MEASUREMENT PROBLEM

Why three scans, and what each equation is for

A radiometer pointed at water does NOT measure water-leaving light. It measures everything arriving along that line of sight, and the water is only part of it:

$$L_t(\lambda) = L_w(\lambda) + \rho L_{sky}(\lambda) + L_{glint}(\lambda)$$

what you want

reflected sky

specular sun

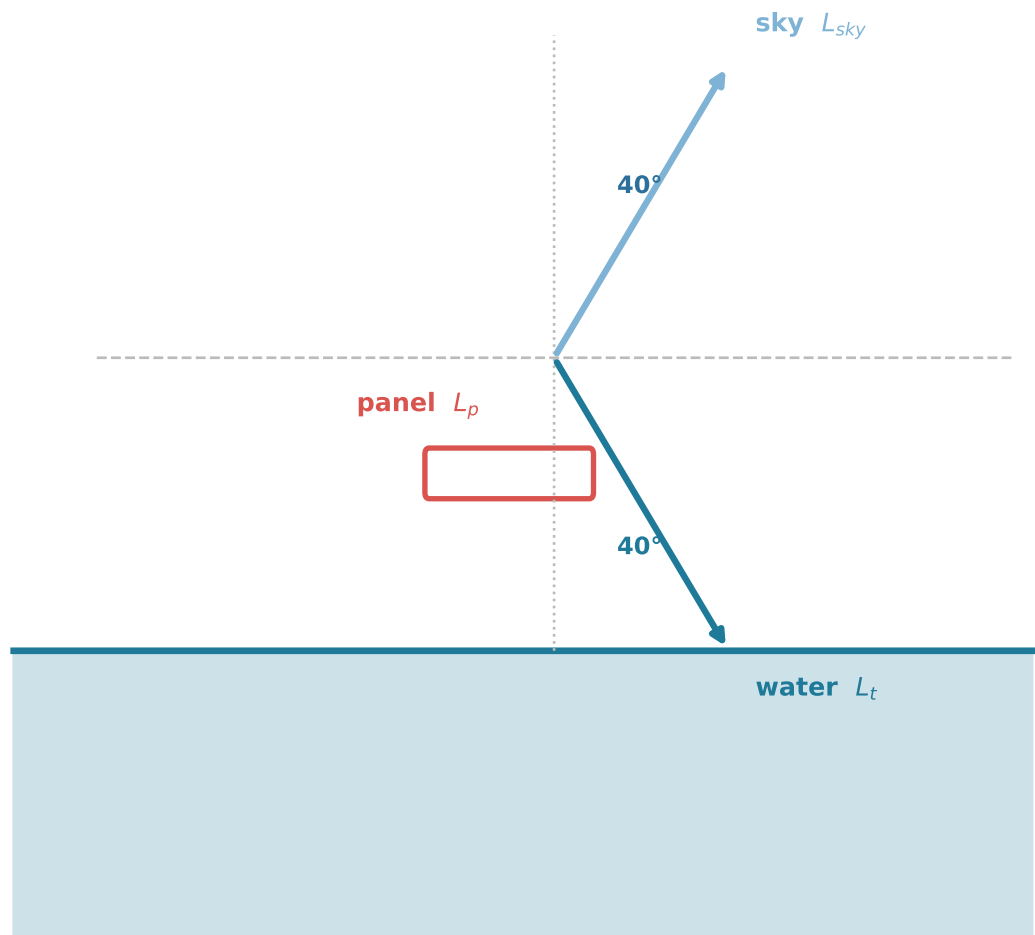
The geometry (40° from nadir, 135° in azimuth from the sun) is chosen to make the third term negligible. The second must be measured and subtracted, which is why there is a sky scan.

THE THREE GOVERNING EQUATIONS

$$E_d(\lambda) = \frac{\pi L_p(\lambda)}{R_p(\lambda)} \quad \text{(F1) panel route to irradiance}$$

$$L_w(\lambda) = L_t(\lambda) - \rho L_{sky}(\lambda) \quad \text{(F2) remove the reflected sky}$$

$$R_{rs}(\lambda) = \frac{L_w(\lambda)}{E_d(\lambda)} - \Delta \quad \text{(F3) normalise, remove residual offset}$$



SKY and WATER are mirrors through the horizontal:
SAME compass bearing, opposite tilt.

ρ — the term everything hinges on

It is NOT the Fresnel coefficient. A flat surface would reflect ~0.021 at 40°. The sea is not flat: it is a distribution of wave facets, each seeing a different part of the sky, so ρ is an EFFECTIVE factor over that distribution.

That is why it depends on wind — wind sets the facet distribution — and why it depends on the sky, because the facets are sampling the sky.

$\rho = 0.028$ (Mobley 1999): 40°/135°, wind < ~5 m/s, clear sky.

KEY CONSEQUENCE: under a UNIFORM sky there is little to sample between, so facet orientation stops mattering and ρ becomes weakly wind-dependent. This is why overcast is workable.

The residual offset Δ (F3)

Even at the best geometry, (F2) leaves a residual from facets that caught the sun, from whitecaps, and from the sky scan not perfectly matching what the surface actually reflected. Three treatments:

none	$\Delta = 0$. The default, deliberately.
nir_zero	assume $R_{rs} = 0$ over 750–800 nm and subtract the mean. VALID in clear oceanic water. In turbid water it DELETES REAL SIGNAL.
nir_similarity	Ruddick et al. (2006): fixed $R_{rs}(780)/R_{rs}(870) = 1.912$ solves for Δ . Usable in turbid water.

Which one is legal is a property of YOUR WATER, not a software preference. That is why the default is 'none': a correction applied in the field to a CSV you keep cannot be undone. Keep the raw .sed.

2 · WHAT YOU HAVE DECIDES WHAT YOU GET

The exe supports both setups. This is which is which.

A · NaturaSpec Plus ALONE

One instrument, narrow field of view (FLENS8 = 8°).
THREE radiance scans, one at a time:

- 1. panel L_p (the REFERENCE scan)
- 2. sky L_{sky}
- 3. water L_t

E_d is INFERRED from the panel by (F1). The panel is a proxy: you never measure E_d , you measure L_p and assume the panel converts it faithfully.

CODE:
`rrs_from_sed(water, sky, source='radiance')`

YOU GET: R_{rs}
YOU CARRY: panel reflectance, panel levelness,
panel-to-target TIME LAG, ρ

B · + SEPARATE IRRADIANCE SENSOR

TWO instruments. The second has a WIDE (hemispherical) field of view and measures E_d directly, logging at the SAME MOMENT as the radiance scans.

- 1. sky L_{sky} } narrow FOV
- 2. water L_t }
- 3. E_d measured, simultaneous, wide FOV

No panel in the calculation at all.

CODE:
`rrs_from_separate_ed(wl, Lt, Lsky, Ed, ed_wl)`

YOU GET: R_{rs} , PAR, atmospheric transmittance,
an E_d -stability check
YOU CARRY: INTER-INSTRUMENT CALIBRATION, ρ

THE TRADE — you are swapping one error for another, not removing error

- A → B REMOVES three errors:

 - the panel-to-target TIME LAG, because E_d is now measured at the same moment as the target. Broken cloud stops being disqualifying.
 - the PANEL REFLECTANCE (your 0.99 assumption), a straight multiplicative bias on every number.
 - PANEL LEVELNESS — the collector defines the horizontal plane itself.
- A → B ADDS one error:

 - R_{rs} divides a radiance from instrument A by an irradiance from instrument B. ANY offset between their absolute calibrations is a DIRECT MULTIPLICATIVE BIAS on every R_{rs} .
Measured: a 6 % gain offset → −5.7 % in $R_{rs}(443)$, uniformly and silently. It does not average out.

THE CROSS-CHECK — keep the panel, but demote it to a TRANSFER STANDARD

Point the radiance sensor at the panel while the irradiance sensor sees the same sky. You now have TWO INDEPENDENT ESTIMATES of the same E_d , and their ratio is the inter-instrument factor:

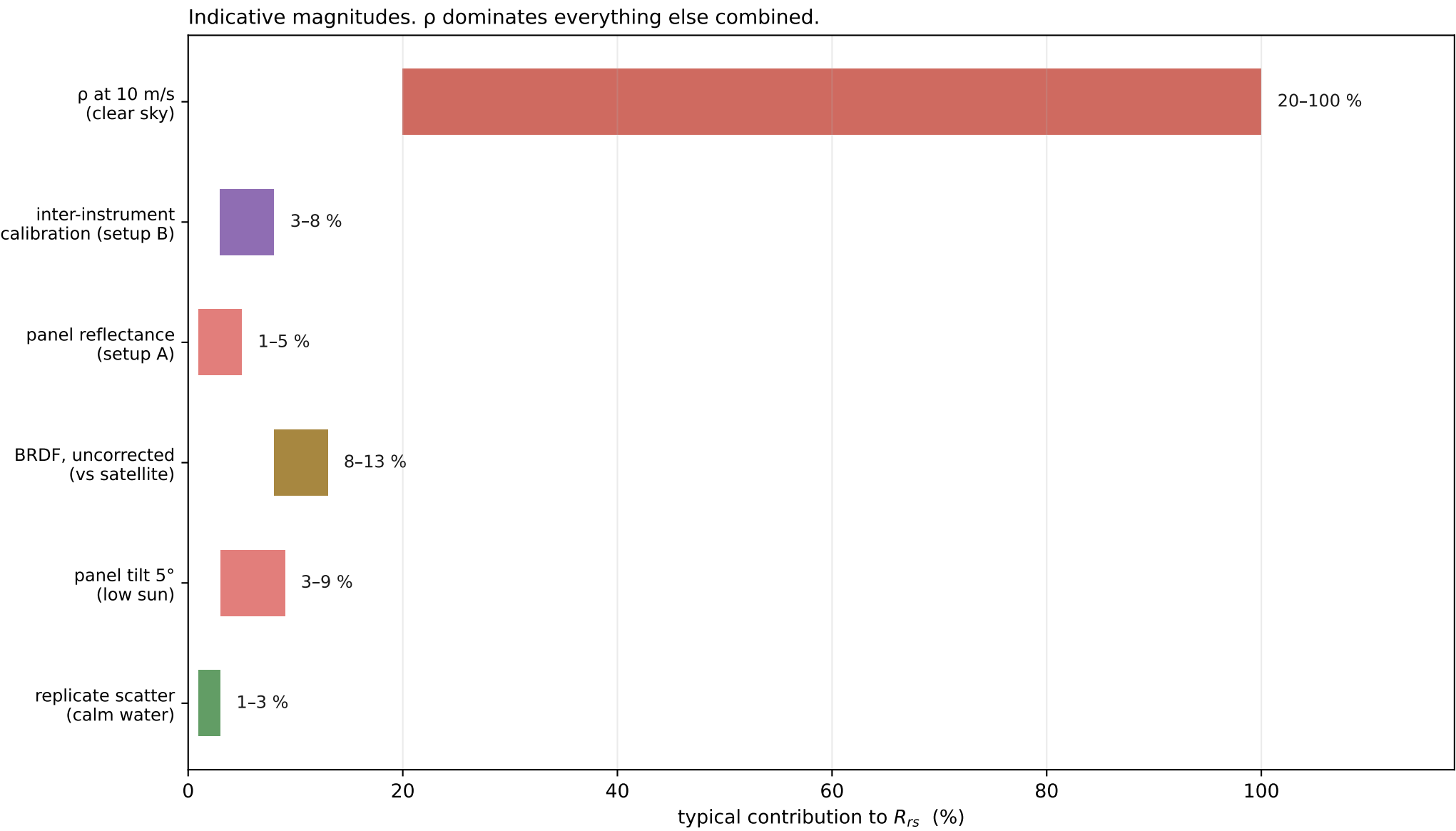
$$C(\lambda) = [\pi \cdot L_p(\lambda) / R_p(\lambda)] / E_d^{measured}(\lambda) \quad (F4)$$

- $C \approx 1$ and FLAT the two instruments agree. Use the measured E_d directly.
- $C \neq 1$ but FLAT a gain offset. Decide which absolute scale you trust before correcting.
- C STRUCTURED in λ NOT a gain. Suspect a stale calibration, a tilted collector, a shaded panel.

Run it ONCE PER DEPLOYMENT, and again after anything gets knocked. Reported by `cross_calibration_factor()`. It is a DIAGNOSTIC first: multiplying C back in re-introduces the panel reflectance that setup B existed to remove.

3 · THE ERROR BUDGET

Which terms actually dominate, with numbers



ρ is the whole game

Its wind dependence comes from wave facets sampling a NON-UNIFORM sky. That single fact explains the whole behaviour:

- clear sky + wind $\rightarrow$ ρ swings widely, and no bundled table covers it (Mobley 2015 is not redistributable). Above ~ 5 m/s the software REFUSES to invent a value.
- uniform overcast $\rightarrow$ little to sample between, so ρ is stable and barely wind-dependent. Overcast is the FRIENDLY case.

Field rule: WHITECAPS APPEARING marks the edge of where $\rho = 0.028$ holds.

What does NOT dominate

Replicate scatter on calm water is 1–3 %. If you are averaging ten scans to chase that, while ρ carries 20–100 %, you are optimising the wrong term.

The useful things to spend effort on, in order:

1. measure and record the WIND
2. get the sky-scan BEARING right (same as the water scan, the commonest field error)
3. cross-calibrate the two instruments once
4. keep the panel clean, dry and LEVEL
5. only then, more replicates

WHAT NONE OF THIS COVERS

These are structural, not statistical, and no amount of averaging touches them:

- R_{rs} is produced AT THE MEASUREMENT GEOMETRY, not normalised to nadir. Satellite products are exact-normalised. The gap is 8–13 % at typical field geometries. Correct with the Morel f/Q ratio (giop-workbench), and note that correction itself omits the air-water transmittance term.
- Under OVERCAST the BRDF correction is not applicable at all: the Morel tables describe a clear-sky field with a direct beam.
- The panel is assumed LAMBERTIAN. Real Spectralon is close but not exact, and worse when dirty.
- Everything assumes the water is HORIZONTALLY UNIFORM over the footprint and the surface is not breaking. Whitecaps in the field of view invalidate the scan outright.

4 · CROSS-CHECKING, AND WHAT TO DO IN THE FIELD

Every check available, and what each one can and cannot catch

1 · E_d STABILITY (needs the irradiance sensor)

Bracket the station with two E_d scans, before and after the target sequence.
ed_stability(before, after, wl) → worst % change + verdict
CATCHES: the light changing mid-station, which violates the core assumption that E_d is the same for reference and target. A 22 % drop reads 'every R_{rs} here is suspect'. With a panel alone you cannot test this without spending another scan, and even then the interval between is unsampled.

2 · INTER-INSTRUMENT $C(\lambda)$ (needs panel + irradiance sensor)

Simultaneous panel scan and E_d scan → equation (F4).
cross_calibration_factor(wl, L_panel, Ed, ed_wl)
CATCHES: a relative calibration offset between the two instruments, which is otherwise a silent multiplicative bias on every R_{rs} . Distinguishes a FLAT disagreement (gain) from a STRUCTURED one (stale calibration, tilt, shading).

3 · ATMOSPHERIC TRANSMITTANCE (needs the irradiance sensor)

$$T(\lambda) = E_d(\lambda) / [F_0(\lambda) \cdot \cos \theta_s] \quad (F5)$$

CATCHES: an objective record of the sky state, replacing 'it looked clear'.
Clear sky 0.6–0.8 in the visible; heavy overcast 0.1–0.3. The SHAPE matters too: clear-sky T rises toward the red as Rayleigh weakens, while cloud is nearly flat, so a FLAT T is itself evidence of cloud. In giop-workbench, which carries the F_0 table.

4 · REPLICATES

Five water and five sky scans per station; the GUI averages and plots them with the individuals.
CATCHES: random surface variability only. It does NOT catch ρ , calibration, or geometry error — those are systematic and shared by every replicate, so the scatter of replicates UNDERSTATES the real uncertainty.

5 · SANITY, on the spot

R_{rs} positive across 400–700? Negative means over-subtraction.
Peak where the water looks? Blue for clear, green for turbid.
 $R_{rs}(443)$ of order 1e-3 to 1e-2?
CATCHES: gross errors, immediately, while you can still re-measure. This is the single highest-value habit: run the GUI AT the station, not at home.

No check here catches ρ . Nothing you can do in the field measures it. That is why the wind speed and the sky state must be RECORDED: they are the only route to a better ρ later.

5 · WHAT THE ABSOLUTE SCALE BUYS

Products a reflectance-only instrument cannot produce

PAR — Photosynthetically Available Radiation

A PHOTON flux, not an energy flux, because photosynthesis counts photons rather than joules:

$$\text{PAR} = \int_{400}^{700} E_d(\lambda) \frac{\lambda}{119.6} d\lambda \quad [\mu\text{mol photons m}^{-2} \text{ s}^{-1}] \quad (\text{F6})$$

with E_d in $\text{W m}^{-2} \text{ nm}^{-1}$ and λ in nm. The constant is $N_A hc = 0.1196 \text{ J} \cdot \text{m/mol}$, expressed for nanometres and micromoles. Nothing here is fitted or tabulated — it is the photon energy hc/λ .

Full midday sun $\approx 2000 \mu\text{mol m}^{-2} \text{ s}^{-1}$. Heavy overcast is one to two orders lower. Under cloud that number IS the record of how much light the water column actually received, which is the context a cloudy-day spectrum needs and which no reflectance ratio can supply.

nLw — normalised water-leaving radiance

The quantity satellite ocean-colour products are actually distributed in:

$$nL_w(\lambda) = R_{rs}(\lambda) F_0(\lambda) \quad (\text{F7})$$

F_0 is the extraterrestrial solar irradiance. This is a pure UNIT CHANGE from R_{rs} , carrying no extra assumption beyond the F_0 spectrum itself, and it is what makes a field spectrum directly comparable with a satellite product.

It does NOT apply the BRDF correction, so the result is still at the measurement geometry, and it applies no Earth–Sun distance correction (F_0 is the mean-distance value, $\pm 3.4 \%$ annually). In giop-workbench, which carries the F_0 table.

FOOTPRINT — what patch of water a scan actually averaged over

From the logged rangefinder distance R and the foreoptic full-angle FOV:

$$\text{across-view} = 2R \tan(\text{FOV}/2) \quad \text{along-view} = \text{across} / \cos \theta_v \quad (\text{F8})$$

$$\text{sensor height} = R \cos \theta_v \quad \text{spot offset from nadir} = R \sin \theta_v$$

YOUR instrument, $R = 3.837 \text{ m}$ with an 8° lens at 40° from nadir: a $0.54 \times 0.70 \text{ m}$ ellipse, 0.30 m^2 , sensor 2.94 m up, spot 2.47 m out.

Three uses: it is the scale wave variability averages over; the 2.47 m offset says whether the footprint clears YOUR OWN SHADOW; and it sets the honest comparison to satellites — an OLCI 300 m pixel is 3.0×10^5 times that area, a MODIS 1 km pixel 3.4×10^6 times.

**All of these need the IRRADIANCE channel, not just a reflectance ratio.
Take an E_d scan at every station, even when the sky is clear.**

6 · WHERE THE SUN HAS TO BE

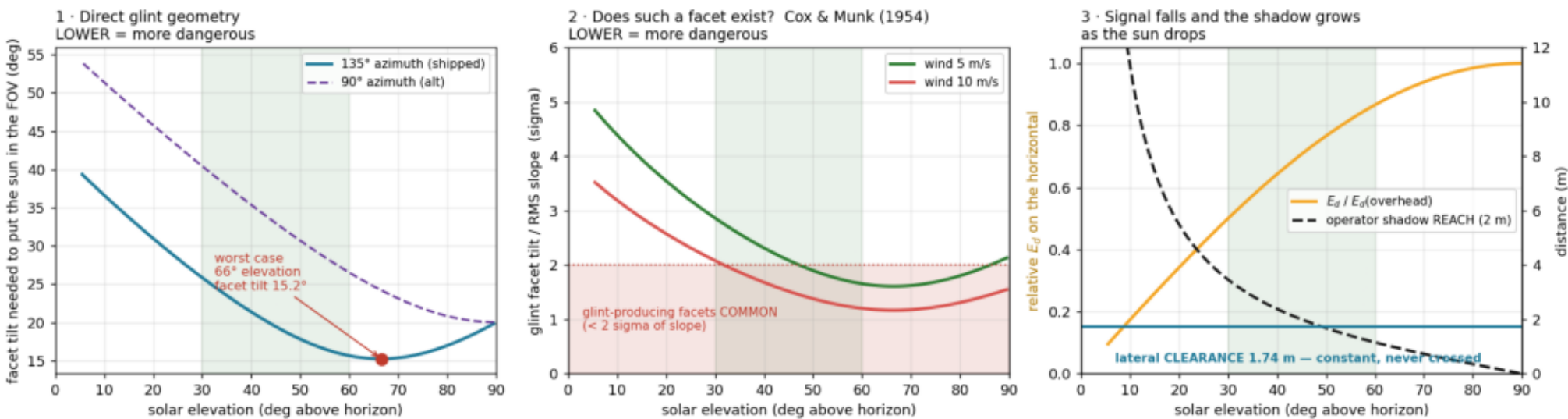
The window, computed rather than adopted

PREFERRED: solar elevation 30° - 60° above the horizon

which is ALSO solar zenith 30°-60° — the window is symmetric about 45°, so it is correct whichever convention you mean.

USABLE 20°-70°. Nothing fails outside 30-60; $\rho = 0.028$ is simply less well determined.

Where the sun has to be — computed, at 40° view zenith / 135° relative azimuth. Green band = elevation 30-60° (= zenith 30-60°, symmetric about 45°). It is the best available COMPROMISE, not a glint-free zone: glint is never eliminated at moderate sun, which is exactly what $\rho = 0.028$ exists to absorb.



TOP end — GLINT, and it is the surprise

A facet puts the sun in the field of view when its normal bisects the sun and view directions. That tilt is MINIMISED at ~66° elevation, at only 15.2°.

Under Cox & Munk (1954) slope statistics ($\sigma^2 = 0.003 + 0.00512 W$) such facets sit 1.6 σ from the mean at 5 m/s — COMMON.

60° → 30° elevation raises the required tilt 15.6° → 25.8° and cuts the glint weight 15× (0.254 → 0.017).

HIGH SUN IS THE GLINT PROBLEM, NOT LOW SUN. Midday is the worst time, not the best.

BOTTOM end — SIGNAL

E_d on the horizontal goes as $\sin(\text{elevation})$:

- elevation 60° → 87 % of overhead
- elevation 30° → 50 %
- elevation 20° → 34 %
- elevation 10° → 17 %

and the airmass roughly doubles from 30° to 20°, so the direct beam is attenuated further and the field grows more DIFFUSE — which also makes the sky scan harder to match to what the surface actually reflected.

NOT a constraint, though everyone assumes it is: YOUR OWN SHADOW

Reach is the wrong test. The shadow runs along the ANTI-SOLAR bearing while the target sits 45° off it at 135° relative azimuth, so the miss distance is a FIXED LATERAL 1.74 m at 3.8 m range — independent of solar elevation. At 10° elevation a 2 m operator throws an 11.3 m shadow and it STILL misses the spot by 1.74 m.

A person never shadows the footprint here. A BOAT or a PIER is a much larger object and can, which is exactly the IOCCG argument for a 90° relative azimuth instead — there the clearance is 2.47 m and the target is well to the side of the structure.

In practice: the good hours are MID-MORNING and MID-AFTERNOON, not noon. Press WHERE IS THE SUN? in the GUI — it prints the elevation, the tier, and which of the two limits you are near.

Reproduce every number on this page: python analysis_solar_window.py